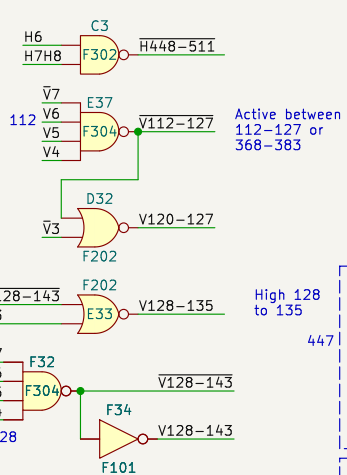
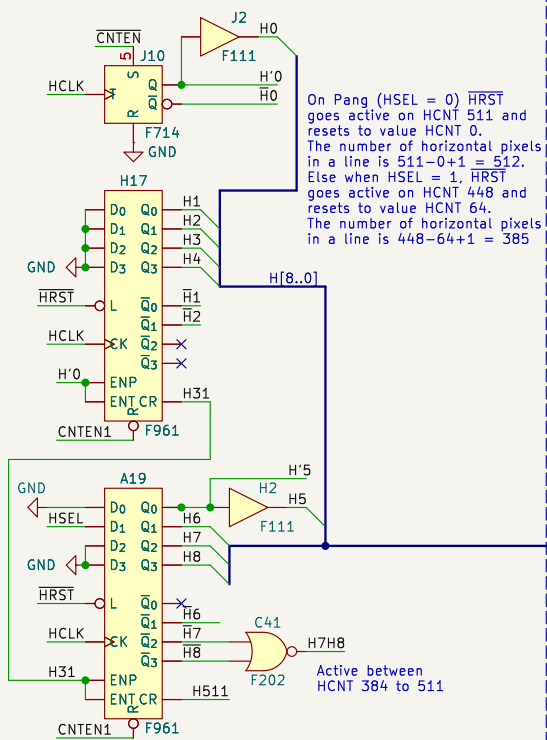
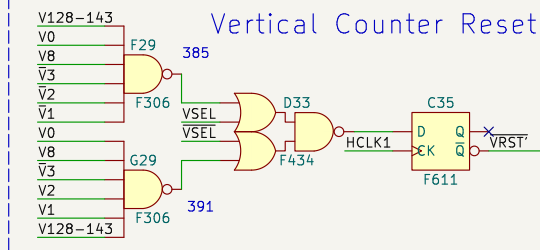


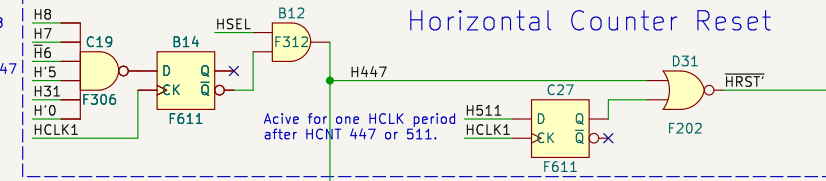
Horizontal Counter



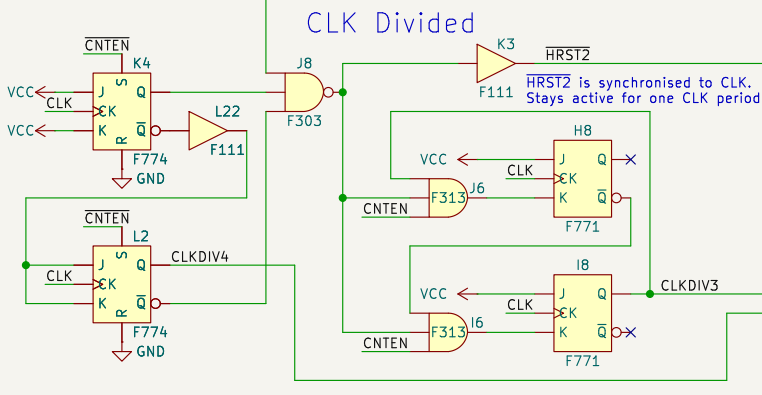
Vertical Counter Reset



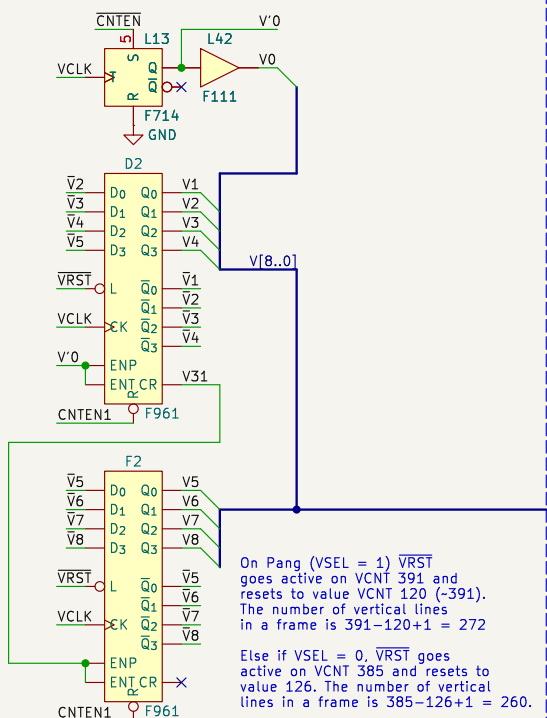
Horizontal Counter Reset



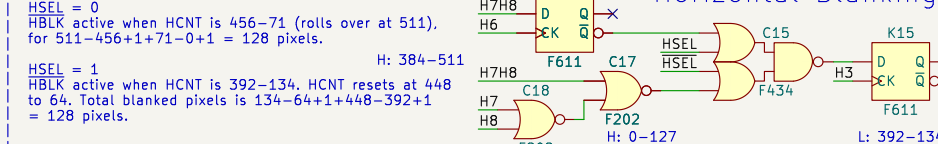
CLK Divided



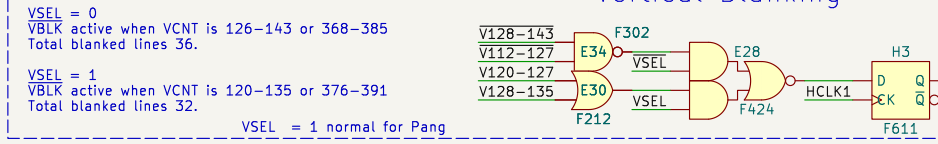
Vertical Counter



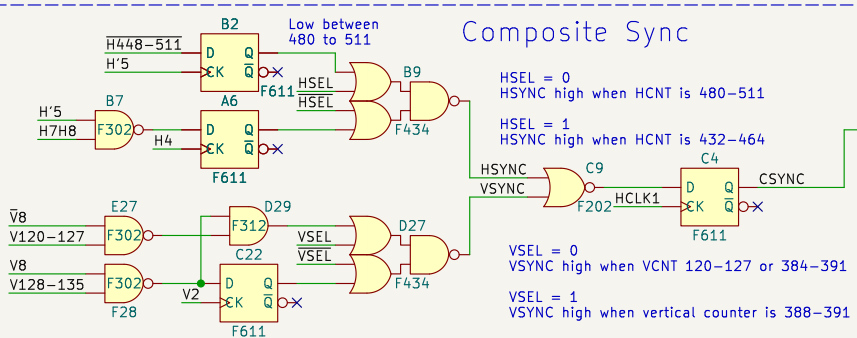
Horizontal Blanking



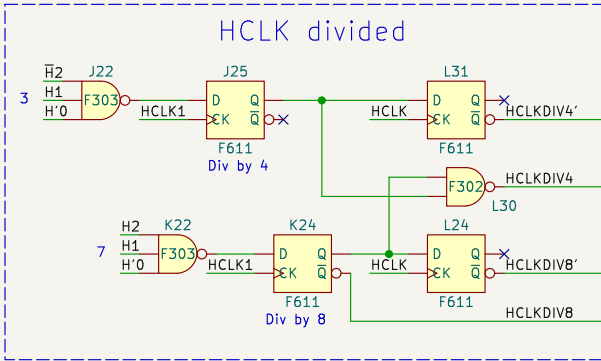
Vertical Blanking



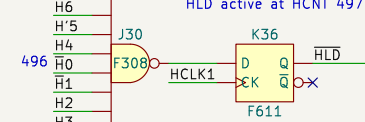
Composite Sync



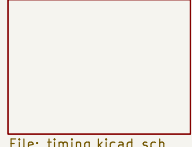
HCLK divided



HLD active at HCNT 497



Timing

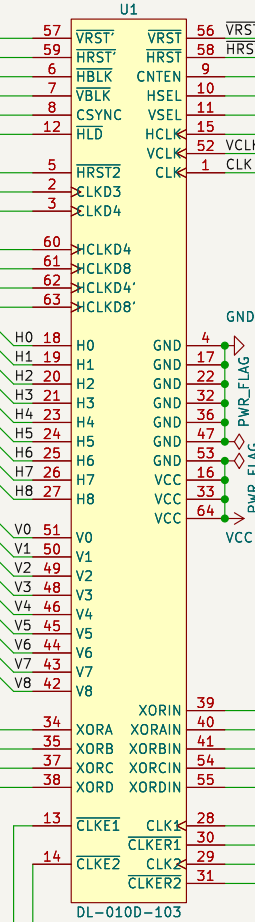


The Schematics have been analyzed from a die picture by InfoSecDJ. The chip is identified as:

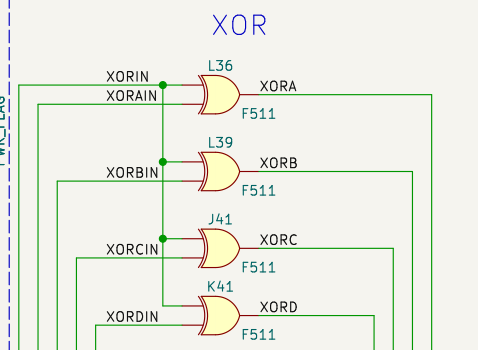
NEC 65006-103

It is a uPD65000 (CMOS-2) series, 3-micron gate array.
- 528 Basic Cells
- 12 Columns, 44 Rows

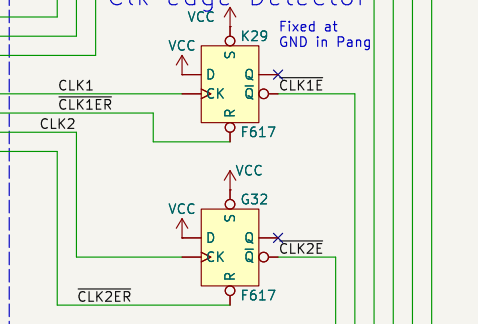
CNTEN Connected to VCC on pang



XOR



Clk edge Detector



Ulf Skutnabba, twitter: @skutis77

Sheet: /
File: dl-010d-103.kicad_sch

Title: **Capcom DL-010D-103**

Size: A3 Date: 2025-09-18
KiCad E.D.A. 8.0.9

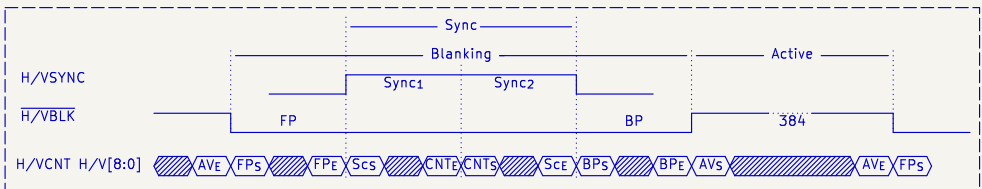
Rev:
Id: 1/2

HSEL	1	0
FP _S ,HBLK _S	392	456
FP _E	431	479
SC _S	432	480
CNT _E	447	511
CNT _S	64	0
SC _E	79	31
BP _S	80	32
BP _E ,HBLK _E	135	71
AV _S	136	72
AV _E	391	455

Horizontal/Vertical Counter Abbreviations

AVS, AVE:	Active Video Start/End
FPS, FPE:	Front Porch Start/End
BPS, BPE:	Back Porch Start/End
CNTS, CNTE:	Horizontal/Vertical Counter Start/Stop
SCS, SCE:	Horizontal/Vertical Sync Start/Stop

Active: Active video signal, $AV = AV_E - AV_S + 1$
 Blanking: Blank video signal
 Sync: Synchronizing video signal
 FP: Front Porch, $FP = FP_E - FP_S + 1$
 BP: Back Porch
 $Sync = Sync_1 + Sync_2$



VSEL	0	1
FP _S , $\overline{\text{VBLK}}_S$	382	376
FP _E	383	387
SC _S	384	388
CNT _E	385	391
CNT _S	126	120
SC _E	127	123
BP _S	128	124
BP _E , $\overline{\text{VBLK}}_E$	143	135
AV _S	144	136
AV _E	381	375

HSEL = 0, VSEL = 1 is used on all games on the Capcom/Mitchell platform.

Total pixels: 512
Active pixels: 384
Blanked pixels: 128
Pixel clock: 8Mhz
Hsync frequency: 15.625kHz
Vertical lines: 272
Blanked lines: 32
Vsync frequency: 57.44485Hz